Amin Maki Digital Design Engineer | FPGA & RTL

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Summary

Electronics Engineering graduate specializing in computer architecture, RTL design, and the RISC-V ecosystem. Hands-on experience in superscalar pipeline optimization (CVA6 core) and hardware verification/synthesis using SystemVerilog, Verilator, Spike, and Vivado. Seeking to pursue graduate studies and research focused on hardware reliability, fault-tolerant systems, and hardware acceleration.

Education

Bachelor of Science in Electronics Engineering, *Shahid Bahonar University of Kerman* 09/2022 – 07/2026
Cumulative GPA: 17.27 / 20.00 (Ranked 2nd among Electronics Engineering Graduates) Kerman, Iran
Bachelor's Thesis: Simulation and Implementation of a Dual-Issue Processor (Score: 20/20)
Selected Coursework: Design of Digital Systems (FPGA & ASIC) (20/20), Microprocessor Systems & Lab (20/20), Computer Programming (20/20), Digital Systems II & Lab (18.2/20, 20/20), Digital Systems I & Lab (17.8/20, 17.6/20)

Skills

RTL Design & Verification
SystemVerilog, Verilog, RTL Design, Testbench Development, Logic Simulation

Computer Architecture
RISC-V ISA, Superscalar Pipelines, Hazard Resolution, Hardware Acceleration

FPGA & EDA Tools
Xilinx Vivado, Verilator, Spike (ISA Simulator), GTKWave, ModelSim, Logic Synthesis

Programming & Software
C/C++, RISC-V Assembly, Bash Scripting, Make/Makefile, Git, Linux

Awards

National Finalist, Scientific Olympiad in Electrical Engineering, *Ministry of Science, Research and Technology* 09/2026

Exceptional Talent Award (Direct M.Sc. Admission Privilege), *Shahid Bahonar University of Kerman* 09/2026

Projects

CVA6-lab , *RISC-V Core Simulation & Custom RTL Modification* 06/2026 – 09/2026
Developed an independent, end-to-end simulation and development framework for the CVA6 RISC-V processor as my BSc Thesis (Grade: 20/20).
Key Achievements & Technical Details:

- Engineered a custom RTL modification to resolve a structural hazard in the write-back stage during multi-cycle execution (e.g., division).
- Implemented a holding buffer and back-pressure logic, yielding up to a +3.57% performance improvement in IPC-sensitive benchmarks without degrading timing (Fmax ~52 MHz) or significantly increasing hardware resources.
- Set up custom C++ testbenches and automated the simulation flow using Verilator, Spike (reference ISA simulator), and Bender.
- Executed full logic synthesis and post-synthesis simulations for 32/64-bit and single/dual-issue configurations using Xilinx Vivado.